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Sustainability criteria for the blue economy

Executive summary

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Executive summary

The blue economy is essential to the future welfare and prosperity of humankind – especially now

At the time of writing, Europe and the rest of the world are slowly entering into the aftermath of the COVID-19 crisis. While it is still too early to assess its impact on the blue economy, the expectation is that the short-term impact on a wide range of traditional activities (notably shipping and tourism) will be severe. Nevertheless, this is a time to reflect upon the recovery and transition path that is ahead of us. The blue economy can and should be part of this trajectory, and this is the moment to reflect upon how to shape its sustainable future.

As a key source of food, energy, minerals, and a means for leisure activities and transportation – upon which hundreds of millions of people depend – the blue economy has the potential to outperform the global economy in terms of both value added and employment growth. The Organisation for Economic Co-operation and Development (OECD) conservatively valued the blue economy's contribution to the global economy in 2010 alone at €1.32 (US $1.5) trillion in gross value added (GVA) terms, with an expected growth to €2.64 (US $3) trillion by 2030 (OECD, 2016)[1]. The blue economy is also essential in reaching the Sustainable Development Goals (SDGs) adopted as part of the Agenda 2030 – in particular SDG 14 "Conserve and sustainably use the oceans, seas and marine resources", but also other indirect goals including responsible production and consumption (SDG 12), affordable green energy (SDG 7) and food to avoid hunger (SDG 2), to name a few.

In shaping the future of the blue economy, it is essential to choose the sustainable pathway

If not well managed and supported, the sector may generate serious damage to global ecosystems and communities. Despite global commitments on the sustainable development of the blue economy, continuing patterns of unsustainable finance support forms of economic development that are not in line with global ambitions. Transition does not come by itself. Investments in oil and gas operations, for example, have important short-term economic and financial benefits – yet they can also have adverse long-term impacts on marine ecosystems and/or local communities. By the same token, certain investments in touristic and cruise operations may provide (short-term) economic benefits to some, while producing negative long-term social and/or environmental externalities to others.

In addition to its potential to drive significant economic and socio-economic development, a sustainable blue economy plays a central role in alleviating the pressures on land resources and fostering climate change mitigation and adaptation. In this context, the European Commission has set out to define a sustainable blue economy and facilitate the establishment of a common understanding at the international level of the activities and practices that contribute to the sustainable development of the blue economy. This study is a step towards the establishment of such a common understanding of sustainable and less sustainable blue economy activities. The overall aims of this study are thus to:

- Define a sustainable blue economy and identify (un)sustainable blue economy activities.
- Develop a blueprint for a BESF, based on sustainability criteria and indicators, which can be applied to blue economy activities and illustrates how environmental, social, economic and governance dimensions of sustainability can be measured within the blue economy sectors.
- Promote the application of such sustainability criteria at the international level.

Towards a working definition of a sustainable blue economy

To develop a working definition of a sustainable blue economy, the study team analysed 32 existing definitions and reviewed these against aspects relating to economic, social, environmental sustainability as well as sustainable governance. Any definition needs to take into account a number of recent trends, including climate change, the uptake of new and innovative technologies and the adoption of circular economy approaches.

[1] OECD (2016), The Ocean Economy in 2030, OECD Publishing, Paris, https://doi.org/10.1787/9789264251724-en.

Based on the elements above, we propose that a sustainable blue economy should include all related marine, ocean-based, coastaland land-based activities that directly or indirectly link to the blue economy. We include all sectors, as this allows currently unsustainable sectors and activities to be captured, which in turn will enable a shift towards more economically, environmentally, and socially sustainable activities that are based on a sustainable governance system that ensures a well-managed and regulated use of ocean resources.

Thus, "*Oceans and seas are essential for the global economy and a healthy planet. Healthy oceans and seas provide jobs, food and drive economic growth. A sustainable blue economy promotes economic growth, social inclusion and improved livelihoods while ensuring the environmental sustainability of the natural capital of the oceans and seas. For the purpose of this report, the sustainable blue economy encompasses all sectoral and cross-sectoral economic activities related to the oceans, seas and coasts. It comprises emerging sectors and economic value based on natural capital and non-market goods and services through the conservation of marine habitats and ecosystem services."*

A total of 30 sustainability frameworks and 500 indicators were identified and prioritised

Of the 30 identified frameworks, 15 were selected for further critical analysis based on the relevance of their criteria and indicators, their reliance on data availability and their degree of transparency, as well as their applicability to different geographical locations, ability to identify risks for unsustainability and potential to address the main threats and trends affecting blue economy sectors. Furthermore, the following sustainability dimensions were considered: the three common pillars of sustainability, being environmental, economic and social sustainability, as well as sustainable governance. Four frameworks were especially relevant for the development of the BESF in the sense that they provided insight on crucial elements for the framework; the four dimensions of sustainability, a cross-sectoral approach for the blue economy and the flexibility for an application at different scales, levels and geographic zones. The four frameworks were analysed in more depth.

In total, more than 500 indicators were identified – most frameworks identified were found to contain suitable indicators, yet not all indicators were related to or referred to specific underlying criteria. Some frameworks made a distinction between core and supplementary indicators: core indicators gather the essential, key or baseline information necessary for a sustainability assessment, whilst supplementary indicators are complementary and more in-depth.

Individually, the existing frameworks do not provide the combination of elements required for the BESF; namely the four dimensions of sustainability, a cross-sectoral approach to the blue economy and the flexibility for an application at different scales, levels and geographic zones. However, the available literature provides for the extraction of strengths from existing frameworks. This approach and methodology has resulted in a coherent BESF that, while not going into the technical (vertical) depths of a specific sector, offers a way to look at multiple sectors simultaneously (horizontally) across a cross-sectoral and harmonized system.

To score or not to score?

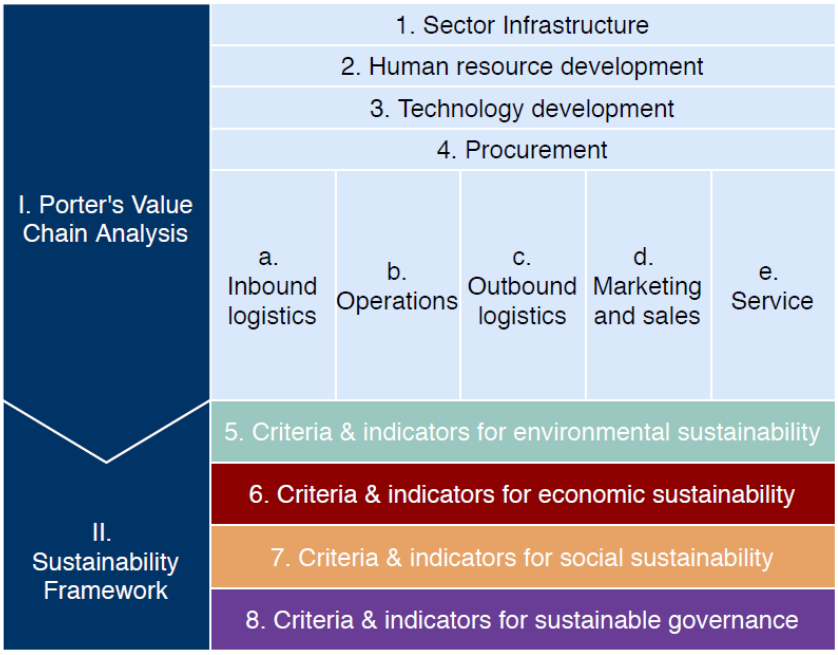
Seven of the analysed frameworks make use of a scoring mechanisms. Scoring mechanisms offer the advantage of allowing for an easy understanding and comparability of indicators. Using a scoring mechanism (or a scale) also allows for an easier application of the framework when data accessibility is limited (e.g. in developing regions), as it allows for less accuracy in the collection of data. However, the application of scales when adequate data is available may also result in indicators that are more imprecise. The reduced precision of scoring approaches can induce difficulties in detecting (incremental) changes. Besides this, a scoring mechanism would make it more difficult to apply the framework at different levels. This is because the performance of a particular subsector will vary according to geographical contexts. Therefore, comparisons cannot be based wholly on the scores of different examples (with varying geographical contexts) that apply the same scoring range. For this reason, a scoring mechanism has not been explored within the context of this study.

The overall approach towards the Blue Economy Sustainability Framework

The structure of the BESF follows a so-called 'nesting' approach: linking criteria and indicators to the four dimensions of sustainability (economic, environmental, social and governance). For each dimension, a number of key sustainability aspects have been clustered into sustainability criteria. Indicators have then been derived from these criteria, allowing for the measurement of key sustainability aspects.

As a conceptual approach, the Value Chain Analysis (VCA) methodology has been adopted for this study. Understanding the interlinkages between the activities of the value chain allows a look beyond the traditional boundaries of a sectoral approach.

Figure 1: The Blue Economy Sustainability Framework approach



Source: own construction, 2020

A set of common criteria and indicators

A selection of common criteria and indicators for the blue economy as a whole is presented in the main report. The BESF contains 44 common indicators, of which 20 are key indicators that determine a set of essential and required minimum criteria and indicators for conducting a review of the sustainability of a given activity. The common criteria retained are the:

- *Environmental dimension*: mitigation, emissions to air, impact on ecosystems, level of energy consumption, energy efficiency and waste/wastewater management;
- *Economic dimension*: concentration of businesses, economic benefits, economic viability, employment, financial viability, public/private funding and costs;
- *Social dimension*: employment conditions, health and safety management, inclusiveness, fairness in remuneration and level of acceptance by stakeholders; and
- *Governance dimension*: permits, impact assessment, innovation, nature-based solutions, risk management, strategy and vision, climate change, certification and labelling, supply chain, level of stakeholder engagement and education on sustainability.

Twelve sets of sector-specific criteria and indicators

Due to the difference in nature between the blue economy sectors, the common criteria and indicators need to be complemented with those that are sector specific. These sector-specific criteria and indicators have been developed for a total of 12 blue economy sectors, in a manner similar to the development of the common indicators; namely, based on the 15 frameworks that were critically analysed for their effectiveness and relevance and that provide a database of over 500 indicators. Sectors for which such criteria and indicators have been developed are: extraction of minerals, extraction of oil and gas, extraction of water (desalination), coastal defence and flood protection, fish and shellfish harvesting, aquaculture, fish and shellfish processing, marine plant and algae harvesting, renewable energy, tourism, transport infrastructure and transport shipping. Data focusing on the environmental, economic, social and governance dimensions established by the BESF have been collected.

Testing the Framework through Case Studies

Through five case studies, the applicability of the BESF has been tested. An example of how the BESF can be applied is provided in the Guidelines which are provided as an accompanying document to this report. The case studies cover a selection of activities and practices across different blue economy sectors. The case studies looked into topics including the following:

- *Aquaculture:* Shrimp farming in the Mekong Delta (Vietnam) and Salmon farming in Norway
- *Extraction of water*: Desalination on Gran Canaria
- *Transport infrastructure*: Planning and operations at the Port of Melbourne, Australia
- *Extraction of Minerals*: Dredging in Harlingen, the Netherlands

Lessons learned in the use of the indicators to collect information for the case studies has contributed towards the development of the BESF as well as the recommendations on its application.

General observations and lessons learned from the case studies

The following points portray the main lessons learned in the process of using the BESF to develop the case studies:

- When applied to the particular example of an activity or a sector, some specific indicators of the framework can be individually interpreted as *stand-alone* (e.g. a very high percentage of *financial returns reinvested in local activities*). However, in most cases it is difficult to come to judgments without having a *benchmark.*
- The indicators of the framework can be applied to different *geographic scales*. The majority of indicators are flexible and can be applied to various scales, subject to data availability. The flexibility of the BESF requires the end-user to clearly identify the scale at which the data on the activity is to be collected, and at which level a sustainability assessment is most appropriate.
- *Addressing externalities*; the VCA has helped to identify and capture potential bottlenecks or externalities (both positive and negative) that provide important context to the data collected through the BESF. For example, stages that occur later in the value chain of the case study activity may provide important incentives for sustainable operations (e.g. the profitability of the activity may significantly depend subsidies or on consumer demand on international markets).
- Attention to detail – *'units and scales';* a challenge may be the identification of data as specified by the unit of the indicator for the correct scale (i.e. national, regional/provincial, company or project level). The BESF, however, provides enough flexibility to use alternative approaches that can alleviate shortcomings in data comparability.
- It is critical to *identify reliable and recent data* to support a sound judgement on the sustainability performance of an activity or sector. More commonly, there was a lack of available data on the social dimension. Other gaps relate to a lack of standardised data collection methodology at the national or international scale.
- Attribution: some BESF indicators may need to be attributed to a particular activity or sector. It can be challenging to identify exactly how much of an aggregated value (e.g. the total budget of a public fund) is dedicated to a particular activity.

- Overly specific indicators can impact the applicability and usability of the framework, and reduce the willingness of stakeholders to engage as data collection becomes too complex. An outcome of this discussion was the development of a set of 20 key criteria and indicators to provide the most essential and required minimum information for conducting a review of the sustainability of a given activity.

- Comparisons between different activities should be conducted on a local basis in order to avoid difficulties in comparing data with geographical, political or environmental differences. Such comparisons could be made, for example, between *different activities* that aim to achieve the same outcome or by looking at the performance of a *single activity* over time. Comparing *operational costs* against the *sustainability benefits* achieved could be useful for such comparisons.

Recommendations on the application of the Blue Economy Sustainability Framework

A set of recommendations has been developed based on the outcomes of the research steps of the study. These recommendations are aimed at economic operators, investors and policymakers at the international level.

- The case studies show that it is important to consider the general and specific context in which an existing or proposed blue economy activity is set.
- The scale of an activity is an important factor. This can relate to the geographic scope (i.e. national, regional, local or company-level) or industry type (i.e. large established industry or new, niche activities).
- Limited data availability or quality should be compensated for by the identification and collection of additional information that the approach used by the BESF allows for.
- The sustainability of an activity needs to be put into the context in which the blue economy activity is implemented, so as to identify critical sustainability needs.
- The BESF requires flexibility over time, to ensure the continued relevance of its sustainability criteria and indicators.
- For a comprehensive set of sustainability data, the collection of information could be standardised and requested by public and private funding institutions.
- A future BESF can support and inform policy levers and actions to secure and promote sustainable investment in the blue economy.

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